CONFIDENTIAL — AMPHORA LABS PROPRIETARY

The Amphora Platform: A Self-Evolving Manufacturing Compiler for Materials

From a neuro-symbolic foundation to autonomous, benchmark-beating discovery

● running today ◐ in active build ○ horizon vision

● SELF-EVOLVING LAYER · EXPANDING

Automated Machine Learning + Autonomous Agents

Amphora-owned, architected by our CTO

MANDATE

Every learned model in the platform is continuously evolved to outperform every benchmark it targets — challengers adopted only on measured, frozen-split wins, written to the audit ledger.

FIXED RULEBOOK

Models and pipelines evolve; judges, gates, and audit instruments stay frozen per release — so self-improvement can never grade itself.

sources & curates data → L0 · invents & upgrades models → L1 · optimizes pipelines & compute → L2 · continuous benchmarking vs published SOTA → all layers

LAYER 3 Programs & Deliverables

● **Naval Maritime Coatings — Huntington Ingalls Industries (MIL-DTL-24441 path)**

- 10-candidate coupled two-part worklist, zero hard-gate violations
- 15-property full target-product-profile scoring
- ~150 °C cured-network glass-transition primer + companion anti-fouling topcoat track
- 2.18× selection yield vs screen-then-discard
- 20 unknown gates preserved fail-closed
- Funded wet-lab gate plan

◐ **Adjacent polymer & coatings programs**

Same compiler, new target-product profiles — reusing judges, gates, and the knowledge graph.

○ **Extreme-environment & adjacent materials classes**

Horizon expansion beyond coatings, on the same trusted foundation.

LAYER 2 Manufacturing Compiler ●

Generate → Score → Filter for Manufacturability → Rank → Audit → Hand-off

FUNNEL: 2,990 candidates → 2,279 buildable → 51 commercial → Top 10

SUPPLY-RESILIENT RE-COMPILATION

Lose a feedstock, re-compile the shortlist, preserve the performance envelope — the clock never restarts.

SCALE

~55 million formulations scored across all 15 properties per day per core; ~10 billion molecules novelty-screened per day at cluster scale; a physical lab tests ~50 per year.

measured results & telemetry →

LAYER 1 Neuro-Symbolic Core

● **NEURAL ENGINE**

- Trained polymer generator (recurrent network, 10.2× diversity gain; conditional diffusion next ○)
- Learned chemistry recognizer (leave-one-family-out validated, 0.969 area-under-curve on hardened benchmark)
- Property-prediction ensembles (gradient-boosted trees + polymer language-model embeddings; glass-transition models exceed published state of the art)
- Out-of-distribution router (tabular foundation model, measured upgrade)
- Continuous embedding search
- Bioisosteric patent-escape operator

◐ **NEURO-SYMBOLIC KNOWLEDGE GRAPH** — materials · feedstocks · process

SYMBOLIC ENGINE ●

- Physics judge bank (cure kinetics, network glass transition, adhesion work, free volume, solubility; molecular-dynamics gates on top leads)
- Fail-closed evidence gates (pass / fail / unknown-fails-closed)
- Coupled Part-A/Part-B stoichiometric projection (1:1 by-volume closure)
- Exact GPU similarity kernel (5.71 billion comparisons/second; full-corpus novelty screen in 1.55 s)
- Synthesis-readiness classifier + route cards
- Freedom-to-operate instrument with live claim-charting

LAYER 0 Data & Knowledge Foundation ●

PI1M polymer corpus (988,775 polymers) · PolyMetriX glass-transition data (7,367 records) · Open Macromolecular Genome (48,213 monomers) · RadonPy molecular-dynamics densities · 51 literature-cited 2D building blocks · Live patent databases (PubChem / SureChEMBL, 116,894-patent genus grounding) · Vendor catalog & pricing feeds · **PROPRIETARY** · Cured-network glass-transition dataset (trade secret, growing) · **PROPRIETARY** · Reverse-engineered incumbent data · **PROPRIETARY** · Program evidence ledgers

● Trust & Audit Spine

CROSS-CUTTING · TOUCHES EVERY LAYER

- Frozen scoring instruments — re-derivable by any skeptic
- Boundary-mode lattice: fixture → locally generated → web-grounded → modeled → production (fails closed)
- Independent grader of a different model class
- Adversarial tamper battery: 12/12 injected cheats caught
- Hash-chained, cryptographically signed audit receipts
- One-command re-derivation of every headline number

THE FLYWHEEL — EVERY PROGRAM COMPOUNDS THE PLATFORM

Customer-funded measurements → Knowledge graph & proprietary datasets → Sharper judges & better models → Faster, cheaper next program ↺